Titan Cloud Gateway: Technical Architecture & Product Manual

Chapter 1: System Overview and Hardware Specifications

Titan Cloud Gateway is an enterprise-grade API gateway designed for high-throughput, low-latency microservice orchestration. Minimum hardware specifications: 4 vCPU cores, 8 GB RAM, and 50 GB NVMe storage. The gateway operates on Linux kernel 5.15 or newer with eBPF network acceleration enabled. It supports up to 50,000 concurrent TLS connections with sub-5 millisecond median latency.

Chapter 2: Authentication Protocols and Rate Limiting

Authentication: Titan Gateway supports OAuth2 with PKCE, JWT validation, mTLS (mutual TLS), and API Key authentication. Tokens must be signed with RS256 or ES256 algorithms. Symmetric HS256 tokens are rejected by default.

Rate Limiting: Uses a distributed Redis-backed sliding window token bucket algorithm. The default global tier limits traffic to 1,000 requests per minute per IP address. Exceeded limits return HTTP status code 429 Too Many Requests.

Chapter 3: Troubleshooting, Diagnostics, and Health Checks

Health Endpoint: The gateway exposes `/healthz` for liveness probes (HTTP 200 OK) and `/readyz` for upstream readiness probes. Metrics: Prometheus metrics are exposed at port 9090 on path `/metrics`. Common Error Codes: Error 5003 indicates upstream backend socket timeout (default 30 seconds). Error 4001 indicates missing or expired JWT authorization header.